SOFTWARE REQUIREMENTS SPECIFICATION

EduShield – Child Education Insurance Management System

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) provides a comprehensive, authoritative description of the EduShield Child Education Insurance Management System. The document serves as the single point of truth for developers, quality assurance engineers, project managers, business analysts, and all relevant stakeholders throughout the system development lifecycle. It establishes a mutual understanding of system behavior, architectural constraints, interface definitions, and non-functional expectations, in full conformance with ISO/IEC/IEEE 29148:2018.

The SRS governs the design, implementation, verification, and validation activities associated with EduShield version 1.0.0. Any feature or behavior not documented herein shall require a formal change request and SRS amendment before implementation.

1.2 Scope

EduShield is a purpose-built, enterprise-grade insurance management platform designed to automate the complete lifecycle of child education insurance policies. The system encompasses every operational phase: initial customer onboarding and application submission, automated risk assessment and underwriting, premium payment tracking, claims processing, compliance auditing, and analytical reporting.

The platform serves four primary user roles – Customer (parent/guardian), Underwriter, Claims Officer, and System Administrator – each operating within a strictly enforced, role-based access control framework. The system replaces manual, paper-based workflows with a fully digital processing pipeline, reducing policy turnaround time from an industry average of several business days to under 24 hours for standard applications.

Core functional modules include:

- User Management and Authentication (JWT and OAuth2)
- Policy Catalog and Configuration
- Child Profile Management
- Policy Application Engine
- Automated Risk Assessment and Premium Calculation
- Subscription Lifecycle Management
- Payment Processing and Financial Ledger
- Claims Management and Payout Scheduling
- Document Management System
- Notification and Communication Engine
- Audit Trail and Compliance Reporting
- Administrative Analytics and Reporting Suite

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification – the present document.
JWT	JSON Web Token – an open standard (RFC 7519) for stateless, cryptographically signed authentication tokens.

RBAC	Role-Based Access Control – an authorization model that restricts system access based on predefined roles.
OAuth2	Open Authorization 2.0 – an industry-standard protocol for delegated authorization via third-party identity providers.
REST	Representational State Transfer – an architectural style for stateless, resource-oriented web services.
API	Application Programming Interface – a defined contract for programmatic communication between software components.
JPA	Java Persistence API – a Java specification for managing relational data via object-relational mapping.
ORM	Object-Relational Mapping – the technique used by Hibernate to map Java entities to database tables.
DTO	Data Transfer Object – a lightweight object used to transfer data between system layers without exposing domain entities.
BCrypt	A password hashing algorithm incorporating salting, used to store user credentials securely.
Premium	The periodic financial payment made by a policyholder to maintain active insurance coverage.
Maturity	The date on which an insurance policy completes its full contractual term and benefits become payable.
Underwriting	The professional process of evaluating applicant risk to determine coverage eligibility and premium loading.
Sum Assured	The guaranteed benefit amount payable to the named beneficiary upon policy maturity or a qualifying claim event.
Grace Period	A defined window (30 days) following a missed premium due date during which the policy remains in force.
Risk Score	A numeric value (0–100) calculated by the Risk Assessment Engine representing the combined risk profile of an applicant.
ACID	Atomicity, Consistency, Isolation, Durability – properties guaranteeing reliable database transaction processing.
CORS	Cross-Origin Resource Sharing – a browser security mechanism controlling permitted cross-domain HTTP requests.
SMTP	Simple Mail Transfer Protocol – the standard protocol for email transmission.
CI/CD	Continuous Integration / Continuous Deployment – automated software build, test, and release pipeline.
WCAG	Web Content Accessibility Guidelines – international standard for web accessibility compliance.
SLA	Service Level Agreement – a contractual commitment defining minimum acceptable system performance thresholds.
DMS	Document Management System – the subsystem responsible for secure document upload, storage, and retrieval.
KPI	Key Performance Indicator – a measurable metric used to evaluate business or system performance.
ISO 29148	International standard specifying requirements for the life cycle processes involved in

	systems and software requirements engineering.

1.4 System Overview

EduShield is implemented as a modular monolithic web application following layered architectural principles. The system comprises a Spring Boot-based backend exposing a RESTful API and an Angular-based single-page application (SPA) frontend. The backend is organized into discrete service domains—authentication, risk assessment, subscription management, claims processing, document management, and notifications—each encapsulating its own business logic while sharing a common data persistence layer backed by PostgreSQL.

Technology Stack Summary:

Layer / Component	Technology
Backend Runtime	Java 17 (LTS), Spring Boot 3.x
Security Framework	Spring Security with JWT (stateless) and Google OAuth2
Data Access Layer	Spring Data JPA with Hibernate ORM
Database (Production)	PostgreSQL 15+
Database (Development / Test)	H2 In-Memory Database
Frontend Framework	Angular 17/18, TypeScript, Tailwind CSS, RxJS
API Documentation	OpenAPI 3.0 (Swagger UI)
Build & Packaging	Maven (backend), Angular CLI (frontend)
Containerization	Docker with multi-stage builds
Orchestration	Kubernetes or standalone VPS (Nginx reverse proxy)
CI/CD Pipeline	GitHub Actions
Email Service	SMTP (configurable provider)
PDF Generation	OpenPDF / iText library
Monitoring	Micrometer with Prometheus integration

1.5 Product Perspective

EduShield operates as a standalone, self-contained web application. While the system is internally self-sufficient, it integrates with the following external services to fulfill operational requirements:

- SMTP Server: External mail relay for transactional email delivery (registration confirmation, OTP, policy certificates, claim status notifications).
- Google OAuth2: Third-party identity provider enabling social sign-in for customers, eliminating the need to manage separate credentials.
- S3-Compatible Cloud Storage: External blob storage for uploaded documents and generated PDFs, with only signed, time-limited URLs exposed via the API.

The system is designed as a direct replacement for manual, paper-based insurance administration processes. It consolidates the functions of multiple legacy workflows into a single integrated platform, reducing policy issuance time from multiple business days to under 24 hours for low-to-medium risk profiles.

1.6 Assumptions and Constraints

1.6.1 Assumptions

- End users have access to a modern web browser (Google Chrome 110+, Mozilla Firefox 110+, Apple Safari 16+, or Microsoft Edge 110+).
- Customers (parents/guardians) possess digital copies of mandatory supporting documents, including birth certificates and income verification records.
- The deployment infrastructure provides a minimum of 4 CPU cores, 8 GB RAM, and 100 GB SSD storage for production.
- The system operates within a jurisdiction that permits electronic signatures and digital document acceptance for insurance purposes.
- An outbound SMTP relay is available and configured prior to production deployment.

1.6.2 Constraints

Constraint Category	Description
Security	All data transmissions must use TLS 1.2 or higher. Passwords must be stored as BCrypt hashes with a minimum work factor of 12.
Availability	The system must maintain a minimum uptime of 99.9% per calendar month, equating to no more than 8.7 hours of unplanned downtime annually.
Regulatory	The system must adhere to applicable data privacy regulations including GDPR and CCPA, as well as local insurance regulatory requirements of the operating jurisdiction.
Data Portability	Development uses H2 in-memory database. The schema and JPA mappings must support zero-modification migration to PostgreSQL for production deployment.
Browser Compatibility	The Angular SPA must render correctly on all evergreen browsers. Internet Explorer is explicitly out of scope.
Financial Controls	Coverage amounts are capped at five times the declared annual income of the applicant as an anti-money laundering and over-insurance control measure.

2. Stakeholder Analysis and User Roles

2.1 User Personas and Role Definitions

The EduShield platform recognizes four primary system actors. Each actor is assigned a distinct role with a well-defined set of permissions, access boundaries, and operational responsibilities. Role assignment is enforced at the API layer via Spring Security and RBAC middleware, ensuring no actor can perform operations outside their designated scope.

Role	System Identifier	Description and Core Responsibilities
Customer	ROLE_USER	The primary end-user of the platform, typically a parent or legal guardian. Responsible for registering an account, maintaining child profiles, submitting policy applications, uploading supporting documents, fulfilling premium payment obligations, and initiating education-related claims against active subscriptions.
Underwriter	ROLE_UNDERWRITER	The risk management professional responsible for reviewing pending policy applications. Analyzes system-generated risk scores in conjunction with uploaded documentation to approve or reject applications. May revise the calculated premium before approval. Accountable for the integrity of the underwriting decision.
Claims Officer	ROLE_CLAIMS_OFFICER	The settlement specialist responsible for verifying claim submissions against policy terms and uploaded evidence. Authorizes or rejects payout disbursements and maintains the integrity of the claims lifecycle. Mandatory rejection reason required for negative decisions.
Administrator	ROLE_ADMIN	The system controller with the highest privilege level. Manages user accounts and staff provisioning, configures policy products, monitors system-wide audit logs, accesses financial analytics, and controls platform health parameters. The Admin role supersedes all other roles in the access hierarchy.

2.2 Responsibility Assignment Matrix (RACI)

The following matrix defines accountability across key system functions. R = Responsible, A = Accountable, C = Consulted, I = Informed.

Function / Feature	Customer	Underwriter	Claims Officer	Administrator
Account Registration and Profile Management	**R**	**–**	**–**	**A**
Child Profile Creation and Maintenance	**R**	**–**	**–**	**I**
Policy Application Submission	**R**	**C**	**–**	**I**
Document Upload and Management	**R**	**C**	**C**	**I**
Risk Score Calculation (Automated)	**–**	**I**	**–**	**A**
Underwriting Decision (Approve / Reject)	**–**	**R**	**–**	**A**
Premium Revision at Underwriting	**–**	**R**	**–**	**A**

Premium Payment Processing	**R**	**I**	**–**	**I**
Claim Submission	**R**	**–**	**C**	**I**
Claim Verification and Review	**–**	**–**	**R**	**A**
Claim Approval and Payout Authorization	**–**	**–**	**R**	**A**
Policy Product Configuration	**–**	**–**	**–**	**R**
User and Staff Account Management	**–**	**–**	**–**	**R**
Audit Log Review	**–**	**–**	**–**	**R**
Financial Analytics and Reporting	**–**	**–**	**–**	**R**
System Health Monitoring	**–**	**–**	**–**	**R**

3. Business Problem and Solution Architecture

3.1 Problem Statement

The child education insurance sector continues to operate with significant operational inefficiencies rooted in legacy, paper-based workflows and fragmented digital systems. These inefficiencies manifest across four principal dimensions:

Problem Area	Impact
High Application Dropout Rate	Complex, non-intuitive legacy interfaces result in customers abandoning applications mid-process, reducing policy conversion rates and increasing acquisition cost.
Delayed Processing Cycles	Manual document verification and risk assessment by underwriting staff can extend policy issuance timelines to several business days, resulting in coverage gaps and customer dissatisfaction.
Lack of Process Transparency	Customers receive inadequate visibility into application status, claim decisions, and rejection rationale, eroding trust and generating high volumes of service inquiries.
Static and Inaccurate Pricing	Uniform premium structures fail to account for individual applicant risk profiles, leading to underpriced high-risk policies and overpriced low-risk ones, creating actuarial imbalance.

3.2 Solution Approach

EduShield addresses each identified problem through a three-pillar solution framework:

3.2.1 Automation

The platform implements end-to-end process automation across all critical workflows. JWT-based authentication eliminates manual session management. The Risk Assessment Engine performs deterministic premium calculation in under 50 milliseconds, bypassing the need for manual actuarial review for standard risk profiles. Automated subscription lifecycle management tracks maturity dates, identifies lapsed policies, and triggers payout processing without human intervention.

3.2.2 Transparency

Role-specific dashboards present real-time, accurate status information at every stage of the policy and claim lifecycle. Customers can track application progress, view payment histories, and receive immediate notification of underwriting decisions. Claims Officers have access to side-by-side comparison views of claimed amounts versus supporting documentation. Audit logs provide administrators with a complete, tamper-evident history of every system event.

3.2.3 Intelligent Risk-Based Pricing

The Risk Assessment Engine replaces static pricing tables with a weighted, multi-factor scoring model that accounts for parent age, occupation risk classification, and income-to-coverage ratio. This ensures that premium rates accurately reflect the actuarial risk profile of each individual applicant, improving portfolio balance and reducing adverse selection.

4. System Features Overview

The following table enumerates all primary system features, their associated functional area, and a brief description of their purpose.

ID	Feature Name	Description
F-01	**Secure Authentication**	JWT-based stateless authentication and Google OAuth2 social sign-in integration for friction-free, secure access.
F-02	**Role-Based Authorization**	Hierarchical RBAC enforced at the Spring Security filter chain level, ensuring strict access segregation between all user roles.
F-03	**Child Profile Management**	Full CRUD operations for child (beneficiary) profiles, including age validation and linkage to policy applications.
F-04	**Policy Catalog**	Browsable catalog of insurance products with configurable parameters: eligibility age range, base premium, coverage amount, and policy duration.
F-05	**Application Wizard**	A state-managed, multi-step frontend wizard guiding customers through the policy application process with document upload integration.
F-06	**Risk Assessment Engine**	Automated, deterministic scoring model that evaluates parent age, occupation, and financial stability to produce a risk score and premium multiplier.
F-07	**Underwriting Workflow**	Structured review interface for underwriters to evaluate applications, view risk scores, revise premiums, and render approval decisions.
F-08	**Subscription Management**	Lifecycle management for active policies, including maturity date calculation, lapse detection, and status transitions.
F-09	**Payment Processing**	Premium payment recording, receipt generation, aggregate payment tracking, and financial ledger maintenance per subscription.
F-10	**Claims Management**	End-to-end claim workflow from customer submission through officer verification to payout authorization and disbursement scheduling.
F-11	**Document Management**	Secure upload, storage, and retrieval of policy-related documents, with mandatory requirement enforcement at the underwriting stage.
F-12	**Notification Engine**	Automated email delivery for all critical system events, utilizing configurable SMTP integration and HTML email templates.
F-13	**Audit Trail**	Comprehensive, tamper-evident logging of all user actions and system state transitions, with paginated retrieval for administrators.
F-14	**Analytics and Reporting**	KPI dashboards, financial yield analysis, risk distribution visualization, and exportable reports in Excel and PDF formats.
F-15	**PDF Certificate Generation**	On-demand generation of policy certificates as secure PDF documents, delivered via email attachment upon policy approval.

5. Functional Requirements

5.1 Authentication and Authorization (FR-01)

The authentication subsystem governs all user access to the EduShield platform. It is implemented using Spring Security and supports two distinct authentication mechanisms: credential-based login and OAuth2 social sign-in.

Attribute	Specification
Requirement ID	FR-01
Feature Name	Authentication and Authorization
Input Mechanisms	Email and password (credential-based); Google OAuth2 token (social sign-in).
Output	A signed JWT Bearer token valid for 24 hours, accompanied by the resolved userId and assigned role.
Password Policy	Minimum 8 characters; must contain at least one uppercase letter, one lowercase letter, one numeric digit, and one special character. Enforced via JSR-303 annotations.
Account Lockout	The account is automatically locked following five consecutive failed authentication attempts. Each lockout event is recorded in the audit log.
Role Hierarchy	ROLE_ADMIN supersedes ROLE_UNDERWRITER and ROLE_CLAIMS_OFFICER; all staff roles supersede ROLE_USER.
Administrative Controls	Administrators may force a password reset or deactivate any user account. Deactivation is a soft operation; account data is retained.
Token Storage	JWT secrets are loaded exclusively from environment variables. Tokens are never persisted in the database.
CORS Policy	Cross-origin requests are restricted to the configured frontend domain in production environments. Wildcard origins are prohibited.

5.2 Policy Management and Configuration (FR-02)

Policy products are defined and maintained exclusively by system administrators. A policy record specifies the commercial and actuarial parameters that govern all applications submitted under that product.

Attribute	Specification
Requirement ID	FR-02
Policy Fields	Policy Name, Minimum Eligible Child Age, Maximum Eligible Child Age, Base Monthly Premium, Risk Coverage Amount, Policy Duration (years), Active/Discontinued status.
Discontinuation Rule	Discontinuing a policy performs a soft delete (status flag update). Existing subscriptions under the policy remain unaffected. No new applications may be submitted against a discontinued policy.
Premium Change Rule	Modifications to the base premium of a policy product affect only new applications submitted after the change date. Active subscriptions retain their original contracted premium.
Age Eligibility	Applications are rejected at submission if the beneficiary child's age falls outside the policy-defined minimum and maximum age range.

5.3 Policy Application Flow (FR-03)

The policy application module provides the primary customer-facing workflow for acquiring insurance coverage. The Angular frontend implements a multi-step state machine wizard, while the Spring Boot backend enforces all business rules and validation constraints.

Attribute	Specification
Requirement ID	FR-03
Date Validation	The policy start date must be a future date. The calculated end date must correspond exactly to the policy's configured duration from the start date.
Income / Coverage Ratio	The selected coverage amount must not exceed five times the applicant's declared annual income. Breaches this constraint return HTTP 422 (ERR_RISK_03).
Document Completeness Check	Applications submitted without all mandatory documents are flagged as DOCUMENTATION_PENDING. The Approve action in the underwriter portal is disabled until all required documents are present.
Duplicate Application Rule	Submitting a second application for the same child under the same active policy is rejected with HTTP 400 (ERR_APP_02). The error response includes a descriptive message.
Application Cancellation	A customer may cancel an application in PENDING status. Cancellation transitions the status to CANCELLED and triggers an audit log entry.
Risk Trigger	Upon application submission, the PolicyApplicationService automatically invokes the RiskAssessmentService. The resulting risk score and category are persisted with the application record.

5.4 Underwriting and Risk Assessment (FR-04)

The Risk Assessment Engine is the actuarial core of EduShield. It implements a deterministic, points-based scoring model that quantifies applicant risk across three dimensions. The computed score is used to categorize the applicant and apply a corresponding premium multiplier. The underwriting decision workflow provides the human oversight layer for non-standard cases.

5.4.1 Risk Scoring Model

Risk Dimension	Criterion	Score (Points)	Rationale
Parent Age	21 – 30 years	5	Youngest cohort; lowest statistical risk.
	31 – 40 years	10	Moderate age; slight elevation.
	41 – 50 years	18	Elevated mortality / morbidity risk.
	51 years and above	25	Highest age-related risk band.
Occupation	Office / Teaching / Engineering	5 – 6	Low occupational hazard.

	Driver / Security / Retail	12	Moderate occupational hazard.
	Construction / Builder	18	High occupational hazard.
	Industrial / Mining	25	Maximum occupational hazard.
Financial Stability	Coverage < 5x Annual Income	5	Comfortably within financial means.
	Coverage 5x – 10x Annual Income	12	Moderate financial stretch.
	Coverage 10x – 15x Annual Income	18	High financial stretch.
	Coverage > 15x Annual Income	24	Exceeds acceptable financial ratio.
Premium-to-Income Load	Monthly Premium > 30% of Monthly Income	+20	Affordability risk indicator.

5.4.2 Risk Category and Premium Multiplier

Risk Category	Score Range	Premium Multiplier	Underwriting Action
LOW	0 – 25 points	1.0x (Standard Rate)	Eligible for automated approval pathway.
MEDIUM	26 – 50 points	1.2x (+20% Loading)	Standard underwriter review required.
HIGH	51 – 75 points	1.5x (+50% Loading)	Enhanced underwriter scrutiny required.
VERY HIGH	76 – 100 points	2.0x (+100% Loading)	Senior underwriter sign-off mandatory; special approval workflow triggered.

5.4.3 Calculation Example

The following example illustrates the premium calculation for a representative applicant:

Parameter	Value	Score Contribution
Base Monthly Premium	$500.00	N/A
Parent Age	45 years (41–50 band)	+18 points
Occupation	Mining	+25 points
Coverage-to-Income Ratio	7x Annual Income	+12 points
Cumulative Risk Score	55 points	Category: HIGH
Applied Multiplier	1.5x	50% Premium Loading
Final Monthly Premium	$750.00	Payable by Customer

5.5 Claims Management (FR-05)

The claims management module handles the complete lifecycle of an education benefit claim from initial submission by the customer through verification by the Claims Officer to final payout disbursement.

Attribute	Specification
Requirement ID	FR-05
Claim Submission	A customer may raise a claim against any subscription in ACTIVE status. The claim must specify the claim type, amount requested, and supporting documentation.
Amount Validation	The claimed amount must not exceed the lower of: (a) the total premium paid to date (totalPaidAmount) or (b) the policy's contracted coverage amount. Violations are rejected with HTTP 422.
Rejection Mandate	A rejection reason is mandatory for all negative claim decisions. Claims Officers cannot submit a rejection without populating the rejectionReason field. This ensures auditability and customer communication compliance.
Payout Schedule	Approved claims are scheduled for payout T+7 business days from the date of Claims Officer sign-off.
Notification	Customers receive email notification upon claim status changes: submission acknowledgment, approval with payout schedule, and rejection with stated reason.
Fraud Detection	Claims Officers have access to a side-by-side verification interface displaying the claimed amount against uploaded supporting documentation. Suspicious patterns are flaggable for escalation.

6. Policy Application Lifecycle (State Machine)

The policy application lifecycle is governed by a formal finite state machine implemented within the PolicyApplicationService. All state transitions are validated against permitted transition rules, and each transition generates a corresponding audit log entry. No state may be bypassed or transitioned to out-of-sequence.

Current State	Triggering Event	Resulting State	Business Rule / Condition
NONE	Customer Submits Application	**PENDING**	System validates input, calculates risk score, persists application record, and notifies underwriters.
PENDING	Underwriter Approves	**APPROVED**	All mandatory documents are present, risk assessment is satisfactory, and underwriter renders affirmative decision.
PENDING	Underwriter Rejects	**REJECTED**	Documentation is insufficient or risk profile exceeds acceptable thresholds. Rejection reason is mandatory.
PENDING	Customer Cancels	**CANCELLED**	Customer voluntarily withdraws the application prior to underwriter action.
APPROVED	First Premium Payment Received	**ACTIVE**	System verifies payment and creates a PolicySubscription record (SUB-XXX identifier). Certificate email is dispatched.
ACTIVE	Grace Period Expires (30 Days Missed)	**LAPSED**	No premium payment recorded within 30 days of the due date. Policy enters LAPSED status. Customer is notified.
ACTIVE	Policy End Date Reached	**EXPIRED**	Policy term has completed. Maturity benefit becomes payable. Subscription record is closed.
ACTIVE	Full Payout Claim Approved	**CLAIMED**	A qualifying claim event results in full benefit disbursement. Policy is closed and subscription record updated.
LAPSED	Reinstatement Payment Received	**ACTIVE**	Customer pays outstanding premium plus reinstatement fee within the reinstatement window. Policy reinstated.
REJECTED	Customer Reapplies	**PENDING**	Customer may submit a new application after addressing deficiencies identified in the rejection.

7. Claim Lifecycle (State Machine)

The claim lifecycle constitutes a distinct state machine operating in parallel with the policy lifecycle. The transitions are governed by the ClaimsManagementService and enforced by the REST API layer. All transitions are audit-logged with actor ID, timestamp, and contextual data.

Current State	Triggering Event	Resulting State	Business Rule / Condition
(New)	Customer Raises Claim	**SUBMITTED**	Claim record created. Validation confirms amount is within permitted limits. Acknowledgment email sent.
SUBMITTED	Evidence Package Uploaded	**SUBMITTED**	Customer uploads supporting documentation. Status remains SUBMITTED pending officer assignment.
SUBMITTED	Claims Officer Assigns	**UNDER_REVIEW**	A Claims Officer accepts the claim for review and initiates document verification.
UNDER_REVIEW	Officer Approves Claim	**APPROVED**	All documentation verified; claimed amount validated against policy terms. Senior sign-off obtained if required.
UNDER_REVIEW	Officer Rejects Claim	**REJECTED**	Discrepancy identified. Rejection reason mandatory. Rejection notification email dispatched to customer.
APPROVED	Finance Schedules Payout	**PAYOUT_SCHEDULED**	System records payout date as T+7 business days from approval date.
PAYOUT_SCHEDULED	Payout Disbursed	**PAID**	Financial disbursement recorded. Subscription totalPaidAmount updated. Customer notified.
REJECTED	Customer Appeals / Resubmits	**SUBMITTED**	Customer may address deficiencies and resubmit within the appeal window defined by policy terms.

8. Database Design and Data Dictionary

The EduShield data model is implemented using Spring Data JPA with Hibernate ORM. The schema is database-agnostic and validated against H2 in development and PostgreSQL in production. The following tables document the primary entities, their attributes, data types, and constraints.

8.1 Entity: users

Column Name	Data Type	Constraints	Description
user_id	BIGINT	PK, IDENTITY, NOT NULL	System-generated surrogate primary key.
email	VARCHAR(100)	UNIQUE, NOT NULL, INDEX	User login credential and primary communication identifier.
password_hash	VARCHAR(255)	NOT NULL	BCrypt hash of the user's password. Plaintext passwords are never stored.
full_name	VARCHAR(150)	NOT NULL	Legal full name of the account holder.
phone_number	VARCHAR(20)	ENCRYPTED	Contact number. Stored encrypted at rest.
role_id	BIGINT	FK → roles.role_id, NOT NULL	References the assigned system role.
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Logical account status. FALSE represents a deactivated account.
failed_login_count	INT	NOT NULL, DEFAULT 0	Counter for consecutive failed login attempts. Resets on successful login.
last_login_at	TIMESTAMP	NULLABLE	Timestamp of the most recent successful authentication.
created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Record creation timestamp.
consent_accepted_at	TIMESTAMP	NULLABLE	Timestamp at which the user accepted Terms and Conditions. Required for GDPR compliance.

8.2 Entity: children

Column Name	Data Type	Constraints	Description
child_id	BIGINT	PK, IDENTITY, NOT NULL	System-generated surrogate primary key.
parent_user_id	BIGINT	FK → users.user_id, NOT NULL	References the parent/guardian account.
full_name	VARCHAR(150)	NOT NULL	Legal full name of the child beneficiary.
date_of_birth	DATE	NOT NULL	Used for age calculation and policy eligibility verification.
gender	VARCHAR(10)	NOT NULL	Gender of the child.

birth_certificate_ref	VARCHAR(500)	NULLABLE	Reference to the stored birth certificate document.
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Logical status. Set to FALSE if child profile is archived.
created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Record creation timestamp.

8.3 Entity: policy_applications

Column Name	Data Type	Constraints	Description
application_id	BIGINT	PK, IDENTITY, NOT NULL	System-generated surrogate primary key.
policy_number	VARCHAR(50)	UNIQUE, NOT NULL	External-facing identifier following the format POL-XXXXXX.
user_id	BIGINT	FK → users.user_id, NOT NULL	References the applying parent account.
child_id	BIGINT	FK → children.child_id, NOT NULL	References the designated beneficiary.
policy_type_id	BIGINT	FK → policy_types.type_id, NOT NULL	References the selected insurance product.
status	VARCHAR(20)	NOT NULL	Current state machine status (PENDING, APPROVED, REJECTED, ACTIVE, LAPSED, EXPIRED, CLAIMED, CANCELLED).
risk_score	INT	NULLABLE	Numeric score (0–100) computed by the Risk Assessment Engine.
risk_category	VARCHAR(20)	NULLABLE	Enumerated risk tier: LOW, MEDIUM, HIGH, or VERY_HIGH.
base_premium	DECIMAL(12,2)	NOT NULL	Base monthly premium at time of application.
final_premium	DECIMAL(12,2)	NULLABLE	Premium after risk multiplier applied and any underwriter revision.
coverage_amount	DECIMAL(15,2)	NOT NULL	Total sum assured under the policy.
start_date	DATE	NOT NULL	Policy commencement date.
end_date	DATE	NOT NULL	Policy maturity date.
underwriter_notes	TEXT	NULLABLE	Free-text notes entered by the reviewing underwriter.
submitted_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Timestamp of initial application submission.
decided_at	TIMESTAMP	NULLABLE	Timestamp of underwriting decision.

8.4 Entity: policy_subscriptions

Column Name	Data Type	Constraints	Description
subscription_id	BIGINT	PK, IDENTITY, NOT NULL	System-generated surrogate primary key.
subscription_number	VARCHAR(50)	UNIQUE, NOT NULL	External-facing identifier following the format SUB-XXXXXX.
application_id	BIGINT	FK → policy_applications.application_id, UNIQUE	One-to-one linkage to the parent application record.
total_paid_amount	DECIMAL(15,2)	NOT NULL, DEFAULT 0	Cumulative sum of all premium payments received against this subscription.
next_due_date	DATE	NOT NULL	Date on which the next premium payment is due.
status	VARCHAR(20)	NOT NULL	Subscription status (ACTIVE, LAPSED, EXPIRED, CLOSED).
maturity_date	DATE	NOT NULL	Calculated date of policy maturity.
created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Subscription creation timestamp.

8.5 Entity: documents

Column Name	Data Type	Constraints	Description
doc_id	BIGINT	PK, IDENTITY, NOT NULL	System-generated surrogate primary key.
application_id	BIGINT	FK → policy_applications.application_id	Associated policy application.
doc_type	VARCHAR(50)	NOT NULL	Document classification (e.g., BIRTH_CERTIFICATE, PARENT_ID, INCOME_TAX_RETURN).
original_filename	VARCHAR(255)	NOT NULL	Original name of the uploaded file as provided by the client.
storage_url	VARCHAR(500)	NOT NULL	Secure storage path or signed URL to the document blob.
mime_type	VARCHAR(100)	NOT NULL	MIME type of the uploaded file (e.g., application/pdf, image/jpeg).
file_size_bytes	BIGINT	NOT NULL	File size in bytes for storage auditing.
uploaded_by	BIGINT	FK → users.user_id	User who performed the upload action.
uploaded_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Timestamp of document upload.

9. API Design and Endpoint Specification

The EduShield backend exposes a RESTful API conforming to the OpenAPI 3.0 specification. All endpoints are documented via Swagger UI. Requests and responses use JSON (application/json) as the content type. All protected endpoints require a valid Authorization: Bearer <JWT> header.

9.1 Authentication Endpoints

Method	Endpoint	Auth Required	Description
POST	/api/auth/register	None	Register a new customer account. Validates email uniqueness and password policy. Sends welcome email.
POST	/api/auth/login	None	Authenticate with email and password. Returns JWT Bearer token on success.
POST	/api/auth/oauth2/google	None	Exchange a Google OAuth2 authorization code for a system JWT. Creates account if first login.
POST	/api/auth/forgot-password	None	Initiate password reset. Sends a 6-digit OTP to the registered email address.
POST	/api/auth/reset-password	OTP Token	Complete password reset using the OTP and new password payload.

9.2 Customer Endpoints

Method	Endpoint	Auth Required	Description
GET	/api/policies	ROLE_USER	Retrieve paginated catalog of active insurance products.
GET	/api/policies/{id}	ROLE_USER	Retrieve detailed specifications for a specific policy product.
POST	/api/children	ROLE_USER	Create a new child (beneficiary) profile.
GET	/api/children	ROLE_USER	Retrieve all child profiles for the authenticated user.
POST	/api/applications/apply	ROLE_USER	Submit a new policy application. Triggers risk calculation. Returns application record.
GET	/api/applications/my	ROLE_USER	Retrieve all applications submitted by the authenticated user.
PATCH	/api/applications/{id}/cancel	ROLE_USER	Cancel a PENDING application. Only applications owned by the authenticated user may be cancelled.
POST	/api/documents/upload/{appId}	ROLE_USER	Upload a supporting document for a specified application. Accepted types: PDF, JPG, PNG.
POST	/api/payments	ROLE_USER	Record a premium payment against an active subscription. Generates payment receipt.
GET	/api/payments/history/{subId}	ROLE_USER	Retrieve complete payment history for a specified subscription.

POST	/api/claims/raise	ROLE_USER	Submit an education benefit claim against an active subscription.
GET	/api/claims/my	ROLE_USER	Retrieve all claims submitted by the authenticated user.

9.3 Underwriter Endpoints

Method	Endpoint	Auth Required	Description
GET	/api/underwriter/dashboard/stats	ROLE_UNDERWRITER	Summary metrics: pending application count, high-risk count, recent approvals.
GET	/api/underwriter/applications	ROLE_UNDERWRITER	Paginated list of applications pending underwriting review, filterable by risk category.
GET	/api/underwriter/applications/{id}	ROLE_UNDERWRITER	Full detail view of a specific application including risk breakdown and uploaded documents.
PATCH	/api/underwriter/applications/{id}/approve	ROLE_UNDERWRITER	Approve an application. Request body may include revisedPremium to override the calculated premium.
PATCH	/api/underwriter/applications/{id}/reject	ROLE_UNDERWRITER	Reject an application. Request body must include rejectionReason (mandatory).

9.4 Claims Officer Endpoints

Method	Endpoint	Auth Required	Description
GET	/api/claims/queue	ROLE_CLAIMS_OFFICER	Paginated queue of claims awaiting review, ordered by submission date.
GET	/api/claims/{id}	ROLE_CLAIMS_OFFICER	Full claim detail view including supporting documents and subscription financial summary.
PATCH	/api/claims/{id}/review	ROLE_CLAIMS_OFFICER	Assign claim to UNDER_REVIEW status, associating it with the acting officer.
PATCH	/api/claims/{id}/approve	ROLE_CLAIMS_OFFICER	Approve claim and schedule payout. System sets disbursement date to T+7 business days.
PATCH	/api/claims/{id}/reject	ROLE_CLAIMS_OFFICER	Reject claim. Request body must include mandatory rejectionReason field.

9.5 Administrator Endpoints

Method	Endpoint	Auth Required	Description
GET	/api/admin/users	ROLE_ADMIN	Paginated list of all user accounts with role and status filters.

POST	/api/admin/users/staff	ROLE_ADMIN	Provision a new staff account (Underwriter or Claims Officer).
PATCH	/api/admin/users/{id}/deactivate	ROLE_ADMIN	Soft-deactivate a user account. Data is retained; access is revoked.
POST	/api/admin/policies	ROLE_ADMIN	Create a new insurance product configuration.
PUT	/api/admin/policies/{id}	ROLE_ADMIN	Update insurance product parameters. Premium changes apply to new applications only.
GET	/api/admin/audit-logs	ROLE_ADMIN	Paginated, filterable audit log retrieval. Supports date range and actor filters.
POST	/api/admin/export/excel	ROLE_ADMIN	Trigger generation and download of an Excel financial report.
GET	/api/admin/analytics/kpis	ROLE_ADMIN	Retrieve current KPI metrics: conversion rate, risk distribution, financial yield.

9.6 Authentication DTO Contracts

9.6.1 Login Request

```
{ "email": "user@example.com", "password": "SecurePass@123" }
```

9.6.2 Login Response (200 OK)

```
{ "token": "eyJhbGciOiJIUzI1NiJ9...", "type": "Bearer", "userId": 42, "role": "CUSTOMER",
"expiresIn": 86400 }
```

9.6.3 Error Response Contract

```
{ "errorCode": "ERR_AUTH_01", "httpStatus": 401, "message": "Invalid credentials provided.",
"timestamp": "2026-03-18T10:00:00Z" }
```

10. Security Architecture and Threat Modeling

10.1 Authentication Security

Security Control	Implementation Detail
JWT Issuance	Tokens are signed using HMAC-SHA256 with a secret key of minimum 256 bits, loaded exclusively from server-side environment variables. Token payload includes userId, role, and expiration claim.
Token Expiration	All issued tokens expire after 24 hours. Refresh token mechanism is available for seamless session extension without re-authentication.
Password Hashing	All passwords are hashed using BCrypt with a minimum work factor of 12 before persistence. Plaintext passwords are never logged, stored, or transmitted.
Account Lockout	Five consecutive failed login attempts trigger automatic account lockout. Lockout events are persisted in the audit log with IP address metadata.
OTP Security	Password reset OTPs are 6 digits, expire after 10 minutes, and are single-use. OTPs are hashed before storage.

10.2 Transport and Communication Security

Security Control	Implementation Detail
TLS Enforcement	All HTTP traffic is served exclusively over TLS 1.2 or TLS 1.3. HTTP-to-HTTPS redirection is enforced at the Nginx layer.
CORS Policy	Cross-Origin Resource Sharing is restricted to the explicitly configured frontend origin in production. Wildcard origin allowance is prohibited.
Security Headers	Nginx is configured to set security headers including: Strict-Transport-Security, X-Content-Type-Options, X-Frame-Options, and Content-Security-Policy.

10.3 Application-Layer Security

Security Control	Implementation Detail
SQL Injection Prevention	Hibernate ORM uses parameterized queries exclusively. No dynamic query string construction is permitted in the repository layer.
Input Validation	All inbound DTO fields are validated using JSR-303 (Jakarta Validation) annotations. Validation errors are captured by the global @ControllerAdvice and returned as structured error responses.
Sensitive Data Encryption	Personally identifiable data fields (phone number, address) are encrypted at rest using AES-256. Encryption and decryption are handled transparently by JPA attribute converters.
File Upload Restrictions	Uploaded documents are validated for MIME type (PDF, JPEG, PNG only) and maximum file size (10 MB). File contents are scanned for executable signatures before storage.
Authorization Enforcement	Spring Security method-level annotations (@PreAuthorize) enforce that users may only access their own records. Cross-user data access is explicitly blocked regardless of role.

10.4 Role Hierarchy

The Spring Security role hierarchy is defined as follows, ensuring that higher-privileged roles inherit all permissions of subordinate roles:

Role	Inherits Permissions Of
ROLE_ADMIN	ROLE_UNDERWRITER, ROLE_CLAIMS_OFFICER, ROLE_USER (full system access)
ROLE_UNDERWRITER	ROLE_USER (plus underwriting-specific endpoints)
ROLE_CLAIMS_OFFICER	ROLE_USER (plus claims-specific endpoints)
ROLE_USER	Base customer access only

11. Frontend Architecture (Angular SPA)

11.1 Core Services

Service	Responsibility
AuthService	Manages JWT token storage in localStorage, provides login/logout methods, exposes an Observable<boolean> for reactive authentication state monitoring across components.
HttpInterceptor	Intercepts all outbound HTTP requests and attaches the Authorization: Bearer <token> header. Also intercepts 401 responses to trigger logout and redirect.
NotificationService	Centralized toast notification system providing success, error, warning, and informational feedback messages across all application modules.
PolicyService	Encapsulates all HTTP interactions with policy-related API endpoints. Implements caching for the product catalog to reduce redundant network requests.
ApplicationService	Manages the application wizard state machine, orchestrates step transitions, persists interim form data, and coordinates document upload operations.
ClaimsService	Handles claim submission, status polling, and evidence upload for the customer-facing claims interface.

11.2 Route Guards

Guard	Behavior
AuthGuard	Prevents navigation to any protected route by unauthenticated users. Redirects to the login page and preserves the attempted URL for post-login redirection.
RoleGuard	Validates that the authenticated user possesses the required role for a given route. Redirects unauthorized users to an access-denied page.
DeactivateGuard	Presents a confirmation dialog when a user attempts to navigate away from the application wizard with unsaved progress, preventing accidental data loss.

11.3 Key UI Components

Component	Module	Description
LoginComponent	Auth	Dual-mode form supporting credential login and Google Sign-In button.
RegisterComponent	Auth	Reactive registration form with 15+ validation rules and real-time feedback.
ApplicationWizardComponent	Customer	Multi-step form using Angular Material MatStepper. Steps: policy selection, child selection, personal details, document upload, review and submit.
ChildListComponent	Customer	Card-based display of registered children with linked policy status indicators.
PolicyTrackerComponent	Customer	Progress visualization for payment history and premium payment timeline.

AssessmentQueueComponent	Underwriter	Filterable data table of pending applications with risk category color coding.
ClaimVerifierComponent	Claims	Side-by-side view of claimed amount versus supporting documentation for validation.
AuditLogComponent	Admin	Paginated, searchable, and exportable audit log viewer.

12. Audit Trail and Notification System

12.1 Audit Log System

EduShield maintains a comprehensive, tamper-evident audit trail for all significant system events. The audit log is immutable; records may only be appended, never modified or deleted. This design ensures a legally defensible compliance record for regulatory review.

Event Category	Logged Events
User Events	Successful login, failed login attempt (with IP), logout, password change, account deactivation, account lockout.
Workflow Events	Application submission, document upload, application cancellation.
Governance Events	Application approval (with final premium), application rejection (with reason), claim approval, claim rejection (with reason), user role change.
Financial Events	Premium payment receipt (amount, subscription reference), payout disbursement, subscription status change (LAPSED, EXPIRED).
System Events	Policy product creation, policy product discontinuation, maturity check execution, scheduled job execution.

Sample audit log record structure:

Timestamp	Actor	Action Code	Entity	Detail
2026-03-18 10:15:04	underwriter-01@co.com	APP_APPROVED	PolicyApplication #452	Premium: 400 → 450
2026-03-18 11:02:33	customer-07@mail.com	CLAIM_SUBMITTED	Claim #88 (Sub-112)	Amount: $1,200

12.2 Email Notification Templates

Template Name	Trigger Event	Content Summary
Welcome Email	Registration Complete	Greets the new user, confirms account activation, provides login link and platform overview.
OTP Verification	Password Reset Request	Delivers a 6-digit OTP valid for 10 minutes with instructions for password reset completion.
Policy Certificate	Application Approved	Multi-part email with attached PDF certificate. Includes policy number, coverage details, and premium schedule.
Application Rejected	Application Rejected	Notifies customer of rejection with the specific reason provided by the underwriter and guidance on reapplication.
Claim Approved	Claim Approved	Confirms claim approval with scheduled payout date and disbursement reference number.
Claim Rejected	Claim Rejected	Communicates rejection reason and appeal procedure to the customer.

Lapse Warning	Payment Overdue (Future)	(Planned) Automated reminder dispatched 7 days before grace period expiry.

13. Data Validation Rules

The following validation rules are enforced server-side via JSR-303 annotations and supplemented by client-side Angular reactive form validators. Server-side validation is the authoritative enforcement layer.

Field	Rule	Constraint	Justification
email	Format	@Email, @NotBlank, UNIQUE	Valid RFC 5321 email address. Uniqueness enforced at database level.
password	Complexity	Min 8 chars, uppercase, lowercase, digit, special char	Compliance with organizational password security policy.
parentAge	Range	@Min(21), @Max(75)	Actuarial underwriting limit; persons outside this range are ineligible.
annualIncome	Minimum	@Min(100000)	Minimum financial stability threshold to qualify for education insurance.
childAge	Maximum	@Max(17)	Education insurance is exclusively available for minors under 18 years of age.
policyDuration	Range	@Min(5), @Max(20)	Minimum and maximum permissible term lengths for education policies.
coverageAmount	Maximum	<= 5x annualIncome	Anti-money laundering control and over-insurance prevention measure.
startDate	Future	@Future	Policy commencement must be a future date.
claimAmount	Maximum	<= totalPaidAmount OR coverageAmount	Claimed amount may not exceed premiums paid or the policy coverage limit.
documentMimeType	Permitted Types	PDF, JPEG, PNG only	File type restriction to prevent upload of executable or malicious files.
documentSize	Maximum	10 MB per file	Storage cost management and abuse prevention.
policyName	Length	@NotBlank, @Size(max=100)	Policy name must be descriptive and within display length constraints.
rejectionReason	Mandatory on Reject	@NotBlank (conditional)	Regulatory requirement: all rejections must carry a documented rationale.

14. Error Codes and Exception Handling

EduShield implements a centralized exception handling strategy via Spring's @ControllerAdvice mechanism. All application exceptions are caught at the controller boundary and mapped to standardized error response objects. The error response contract is consistent across all endpoints.

Error Code	HTTP Status	Description	Cause	Recovery Action
ERR_AUTH_01	401	Invalid Credentials	Email or password does not match any active account.	Redirect to login; display error message.
ERR_AUTH_02	401	Token Expired	Submitted JWT has exceeded its 24-hour validity window.	Prompt user to re-authenticate.
ERR_AUTH_03	403	Access Denied	Authenticated user lacks the required role for the requested resource.	Display access-denied page; do not expose endpoint details.
ERR_AUTH_04	423	Account Locked	User account locked after five consecutive failed login attempts.	Advise password reset via OTP flow.
ERR_APP_01	404	Application Not Found	No application record exists for the supplied identifier.	Verify application ID and retry.
ERR_APP_02	400	Duplicate Application	An active application already exists for this child under this policy.	Display informational modal; offer view of existing application.
ERR_APP_03	400	Invalid Application Status	Requested operation is not permitted for the application's current state.	Refresh application status and re-evaluate available actions.
ERR_RISK_03	422	Insufficient Income	Coverage amount requested exceeds five times the declared annual income.	Suggest reduced coverage amount or alternative product.
ERR_DOC_04	415	Invalid File Type	Uploaded file type is not in the permitted list (PDF, JPEG, PNG).	Request re-upload in a permitted format.
ERR_DOC_05	413	File Size Exceeded	Uploaded file exceeds the 10 MB maximum size limit.	Request compressed or reduced-resolution re-upload.
ERR_CLM_01	422	Claim Amount Exceeded	Claimed amount exceeds permitted maximum based on policy terms.	Revise claim amount and resubmit.
ERR_CLM_02	400	Missing Rejection Reason	Rejection decision submitted without a mandatory rejection reason.	Populate rejectionReason field and resubmit decision.

ERR_SYS_01	500	Internal Server Error	Unhandled exception in application processing layer.	Log error reference; retry after 30 seconds.
ERR_SYS_02	503	Database Timeout	Database connection pool exhausted or query timeout exceeded.	Automatic retry with exponential backoff; escalate if persistent.

15. Non-Functional Requirements

15.1 Performance

Metric	Target SLA	Measurement Approach
Authentication Response Time	< 150 ms (p95)	JMeter load test under 200 concurrent sessions.
Risk Score Calculation	< 50 ms (p99)	Micrometer timer on RiskAssessmentService.calculate().
PDF Certificate Generation	< 2,000 ms (p95)	Spring StopWatch instrumentation on PdfService.
API Response Time (General)	< 300 ms (p95)	Prometheus metrics scraped and visualized in Grafana.
Page Load (TTFB)	< 1,200 ms on 4G	Lighthouse performance audit on Angular production build.
Concurrent User Capacity	1,000+ concurrent sessions	LoadRunner scenario test on production-equivalent infrastructure.

15.2 Reliability and Availability

Requirement	Specification
Target Uptime	99.9% per calendar month. Planned maintenance windows must be communicated 48 hours in advance and may not exceed 4 hours.
Data Durability	Daily automated PostgreSQL snapshots retained for 30 days. Point-in-time recovery (PITR) enabled.
Transaction Integrity	All financial operations are wrapped in database transactions with ACID guarantees. A database restart must not result in data loss.
Load Balancing	Nginx configured as a reverse proxy with upstream load balancing across a minimum of two Spring Boot application instances.

15.3 Maintainability and Code Quality

Requirement	Specification
Layered Architecture	Strict separation between Controller, Service, Repository, and DTO layers. Cross-layer dependency injection is managed by Spring IoC.
Code Coverage	Minimum 70% unit test coverage for all service layer classes, enforced by Maven Surefire and JaCoCo reporting.
Static Analysis	SonarQube integration with zero tolerated Blocker or Critical severity issues at the point of release.
API Documentation	Every public API endpoint must be annotated with OpenAPI 3.0 @Operation, @ApiResponse, and @Parameter annotations.
Logging	SLF4J with Logback. Daily rolling log files retained for 14 days. Sensitive data (passwords, tokens) must never appear in log output.

Versioning	Semantic Versioning (SemVer 2.0.0) applied to all releases. API versions prefixed in the URL path (e.g., /api/v1/).

15.4 Accessibility (WCAG 2.1 Compliance)

Criterion	Requirement
Color Contrast	Minimum contrast ratio of 4.5:1 for all text elements against their backgrounds, as per WCAG 2.1 Level AA.
Keyboard Navigation	All interactive elements (forms, buttons, modals, data tables) are fully navigable using keyboard controls alone.
Screen Reader Support	All dynamic charts, data tables, status indicators, and interactive buttons carry appropriate ARIA labels and roles.
Responsive Design	Fluid layout supporting viewport widths from 320 px (mobile) to 2560 px (ultra-wide desktop). No horizontal scrolling on supported viewports.

16. Deployment Architecture

16.1 Infrastructure Stack

Component	Specification
Operating System	Ubuntu Server 22.04 LTS (64-bit)
Web / Proxy Server	Nginx 1.24+ configured as a reverse proxy and TLS termination point.
Application Server	Spring Boot embedded Tomcat, packaged as an executable JAR. Minimum 2 instances behind Nginx upstream.
Database Server	Managed PostgreSQL 15+ instance with connection pooling (HikariCP), read replica, and daily automated snapshots.
Containerization	Docker with multi-stage builds. Separate images for backend and frontend (Nginx-served Angular build).
Orchestration	Kubernetes cluster (recommended for production scale) or standalone Docker Compose for smaller deployments.
CI/CD Pipeline	GitHub Actions: automated build, test, Docker image push, and deployment on merge to main branch.
Monitoring	Prometheus metrics collection from Spring Boot Actuator. Grafana dashboards for visualization. Alertmanager for threshold breach notifications.
Log Management	Logback output aggregated via a log shipper (e.g., Filebeat) to a centralized log store for long-term retention and search.

17. Testing Strategy

Test Level	Tooling	Scope and Objectives
Unit Testing	JUnit 5, Mockito	Individual service methods and utility classes in isolation. Coverage target: minimum 70% line coverage on service layer.
Integration Testing	@SpringBootTest, H2, MockMvc	API endpoint routing, request validation, database persistence, and service interaction using an in-memory H2 database.
System Testing	Postman Collection, Selenium	End-to-end scenario testing covering full workflow paths: registration through claim payout. Includes edge case and negative path testing.
Performance Testing	JMeter, LoadRunner	Load testing to validate SLA targets under concurrent user load. Stress testing to identify system breaking point.
Security Testing	OWASP ZAP, SonarQube	OWASP Top 10 vulnerability scanning. Static code analysis for security anti-patterns.
User Acceptance Testing	Manual, Stakeholder-led	Business workflow validation by representative users. UI/UX clarity and responsiveness verification across supported browsers.

18. Document Management System

18.1 Policy Document Requirements

Each insurance product type defines a specific set of mandatory and conditional document requirements, stored in the policy_document_requirements configuration table. The underwriter portal enforces these requirements by disabling the approval action until all mandatory documents for the selected policy type are present.

Document Type	Requirement Level	Applicability Condition
Child Birth Certificate	Mandatory	All policy types.
Parent / Guardian Government-Issued ID	Mandatory	All policy types.
Most Recent Income Tax Return	Mandatory	All policy types. Used for income verification and coverage ratio calculation.
School / Educational Institution Enrollment Certificate	Mandatory	Required for claim submission; not required at application stage.
Audited Financial Statement	Conditional	Required when coverage amount exceeds $1,000,000.
Medical Fitness Certificate	Conditional	Required when parent risk category is VERY_HIGH.

18.2 Storage Architecture

Layer	Implementation
Metadata Storage	Document metadata (id, type, original filename, upload timestamp, uploader reference) is stored in the documents table within the primary PostgreSQL database.
Blob Storage	Binary document content is stored in an S3-compatible object storage bucket. Only time-limited, signed URLs are exposed via the API, preventing unauthorized direct access.
Access Control	Document retrieval API enforces ownership validation. Customers may only retrieve documents associated with their own applications. Staff access is governed by role.

19. Compliance and Data Privacy

Compliance Area	Implementation Measure
Right to be Forgotten (GDPR Art. 17)	User deactivation flow anonymizes personal identifiers (name, email, phone) while retaining anonymized financial records for regulatory audit requirements.
Encryption at Rest	Sensitive personal data fields (phone number, home address) are encrypted at the JPA attribute level using AES-256 before persistence.
Encryption in Transit	TLS 1.2+ enforced at the Nginx layer for all client-server communications.
Consent Tracking	Explicit acceptance of Terms and Conditions is recorded at the point of registration, including timestamp. Required for GDPR and CCPA compliance.
Data Residency	All customer data is stored within the regional jurisdiction of the licensed insurance carrier operating the platform.
Audit Immutability	Audit log records are append-only and protected by database-level constraints preventing modification or deletion.
Anti-Money Laundering	Coverage amount capped at five times declared annual income. Financial transactions above regulatory thresholds generate compliance flags for administrator review.

20. Analytics and Reporting Suite

20.1 Key Performance Indicators

KPI	Definition and Calculation
Application Conversion Rate	Percentage of submitted applications that transition to ACTIVE subscription status. (ACTIVE count / Total Submitted) x 100.
Risk Distribution	Percentage breakdown of current active portfolio by risk category (LOW / MEDIUM / HIGH / VERY_HIGH). Visualized as a donut chart.
Financial Yield	Ratio of total premium revenue collected to total claim payouts disbursed over a specified period. A yield below 1.0 indicates net outflow.
Claim Approval Rate	Percentage of submitted claims that result in an APPROVED outcome within a reporting period.
Average Processing Time	Mean elapsed time (in hours) between application submission and underwriting decision, tracked separately for approval and rejection pathways.
User Growth Rate	Month-over-month percentage change in new customer registrations and active subscriptions.

20.2 Export Capabilities

Export Type	Format	Content
Financial Report	Excel (.xlsx)	Policyholder list with premium status, active subscriptions, claims summary, and financial totals for the selected period.
Payment Receipt	PDF	Itemized payment confirmation generated on demand per premium payment.
Audit Trail Export	CSV	Complete system audit log for a specified date range. Suitable for regulatory submission.
Policy Certificate	PDF	Signed policy certificate generated via OpenPDF at the point of subscription activation.

21. Future Scope and Development Roadmap

Phase	Feature	Description
v1.1	AI-Assisted Underwriting	Integration of a machine learning model trained on historical application outcomes to provide predictive claim probability scores alongside the deterministic risk engine.
v1.2	Mobile Wallet Integration	Premium payment via Google Pay and Apple Pay, reducing payment friction for mobile users and improving on-time payment rates.
v1.3	Multilingual Support	Localization of the Angular frontend to support Hindi and Spanish, enabling deployment in India and Latin American markets.
v1.4	Premium Payment Reminders	Automated scheduled job dispatching premium due date reminders via email and (optionally) SMS 7 days before and on the due date.
v2.0	Mobile Application	Native iOS and Android applications exposing the full customer-facing workflow, built on a shared Angular Ionic framework.
v2.1	Biometric Authentication	Integration of device biometric authentication (fingerprint, Face ID) for the mobile application as an alternative to password entry.

22. Glossary

Term	Definition
Actuary	A business professional specializing in the measurement and management of risk and financial uncertainty using mathematics and statistical methods.
Beneficiary	The child in whose name the education insurance policy is issued, and who is entitled to receive the policy benefit upon maturity or a qualifying event.
Coverage Amount	The total sum assured under the insurance policy, representing the maximum financial benefit payable to the beneficiary.
Grace Period	A contractually defined period (30 days in EduShield) following a missed premium due date during which the policy remains in force and coverage is maintained.
Lapse	The termination of insurance coverage resulting from the policyholder's failure to pay the required premium within the grace period.
Maturity	The date on which the insurance policy completes its full contractual term and the sum assured becomes payable to the beneficiary.
Policy Number	The unique external identifier assigned to each policy application, following the format POL-XXXXXX.
Premium	The periodic financial payment made by the policyholder to the insurer in exchange for maintaining active insurance coverage.
Reinstatement	The restoration of a lapsed policy to active status upon payment of all overdue premiums and applicable reinstatement fees within the permitted window.
Risk Loading	The additional premium charged above the standard base rate to compensate for elevated actuarial risk identified during underwriting.
Subscription	The active policy record created upon first premium payment, uniquely identified by a SUB-XXXXXX reference number.
Sum Assured	Synonymous with Coverage Amount. The guaranteed benefit amount payable upon policy maturity or a qualifying claim event.
Underwriting	The professional process by which an underwriter evaluates the risk of insuring an applicant and determines appropriate coverage terms and premium.

23. Conclusion

This Software Requirements Specification documents the complete functional, architectural, and non-functional requirements for the EduShield Child Education Insurance Management System, version 1.0.0. The document has been prepared in strict conformance with the ISO/IEC/IEEE 29148:2018 standard for systems and software requirements engineering.

EduShield represents a comprehensive, enterprise-grade response to the operational deficiencies that characterize the child education insurance sector. By automating the policy lifecycle through a deterministic risk engine, structured underwriting workflows, and a rule-governed claims management system, the platform reduces issuance timelines from days to hours while maintaining the actuarial rigor and compliance standards required of regulated insurance operations.

The system architecture—built on Java 17, Spring Boot 3.x, Angular 17/18, and PostgreSQL—provides a stable, scalable, and secure foundation for enterprise deployment. The modular service design, comprehensive audit trail, and robust security architecture collectively ensure that EduShield is not only functionally capable but also operationally resilient and regulatory-compliant.

All subsequent development activity, feature enhancements, and system modifications must be reconciled with this specification. Changes of sufficient scope require a formal SRS amendment, reviewed and approved by the project architect and relevant stakeholders prior to implementation.

[DOCUMENT SEALED] – SRS v1.0.0 – EduShield Child Education Insurance Management System

ISO/IEC/IEEE 29148:2018 Compliant | Confidential | March 18, 2026